ENGINEERING BRIEF · PROBLEM DEFINITION · PASS 1 OF 2

Two jobs, one database.

Our storefront and our analytics are competing for the same system. This brief defines that problem — who it hurts, what "fixed" must prove, and what we are *not* solving yet. It deliberately names no technology; the solution comes in Pass 2.

PREPARED BY	PURPOSE	SCOPE	DATE
Data Engineering	Team consensus on the problem	Problem, not solution	June 20, 2026

THREE VITALS, THE WAY THEY READ TODAY → THE WAY THEY MUST READ

Time to answer an analytical question	Engineering capacity spent on upkeep	Age of the data we decide on
12 min → <5 sec	60–70% → 20–30%	24 hr → <5 min

engineering-brief.pdf

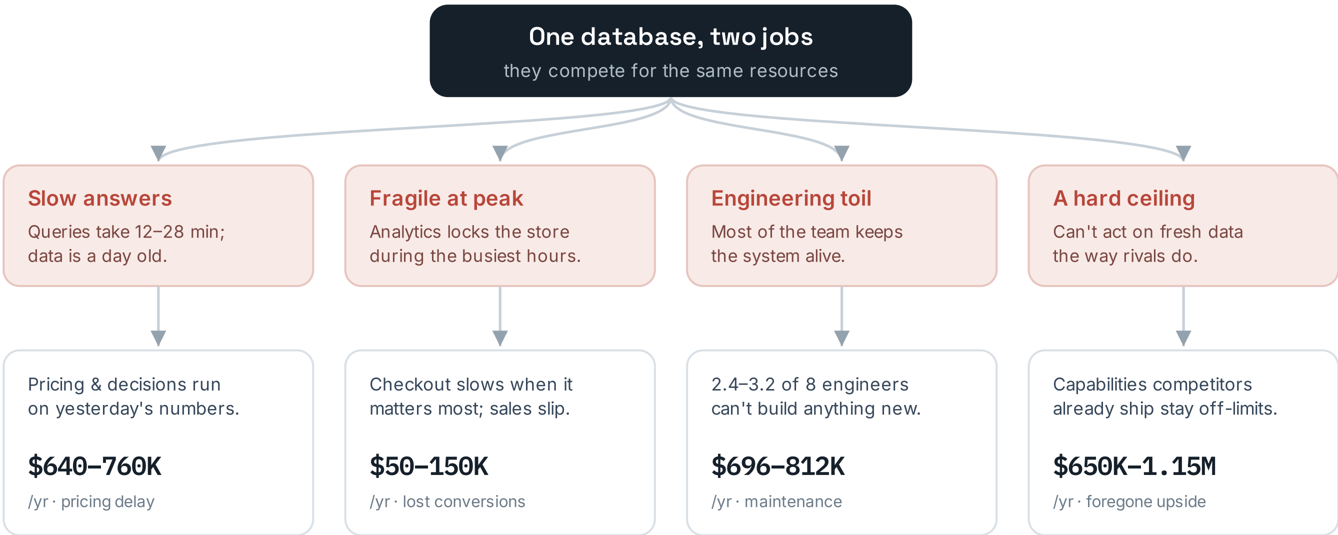
Internal · Confidential · For review and alignment

01 The problem, in plain terms

root cause

One database is doing two incompatible jobs at once: **running the store** (processing ~50,000 orders a day) and **understanding the store** (the reports and analysis the business runs on). These two jobs want opposite things from the same machine, so they fight over it.

That single unresolved conflict is the root cause. Everything below — slow answers, a storefront that wobbles at peak, an engineering team stuck on maintenance, and a ceiling on what we can build next — is a downstream symptom of it. The fix is not a faster machine or a smarter query; it is giving each job its own place to run so they stop competing.



COST OF LEAVING IT ALONE

≈ $2.08M / yr run cost, status quo

$7.6M over three years — the baseline every option is judged against.

FIG. 1 — HOW ONE ARCHITECTURAL CONFLICT CASCADES INTO FOUR BUSINESS PAINS. FIGURES ARE THE BRD'S OWN ESTIMATES.

02 Who's hurting, and how

impact map

The pain is not evenly spread. Five groups feel it in different currencies — time, reliability, revenue, and morale.

The data team DEEPEST WOUND

Roughly 5 of 8 engineers spend their time firefighting a misused database instead of building. Incidents take a mean of 5.2 hours to resolve.

60–70% of capacity on upkeep · 2.8 incidents/mo

Customers & revenue WORST TIMING

During Black Friday and Cyber Monday, heavy analytics locks tables and slows checkout — degrading sales in the exact windows that matter most.

0.1–0.3% conversion lost at peak · ~$50–150K/yr

Pricing & product DECISION LAG

Pricing optimization runs once a day on overnight batch. Competitors with fresh data adjust 24–96 times in a single peak window; we manage 1–2.

1×/day vs. intraday rivals · ~$640–760K/yr

Analytics consumers SLOW & STALE

The ~50 people on ~30 dashboards wait minutes for each answer, and the answer describes the business as it was up to a day ago.

12–28 min per query · on 24-hour-old data

Finance & the business SLOW BLEED

Today's setup costs ~$2.08M a year just to stand still, and the compounding cost of inaction — delay, fragility, toil, and foregone capability — reaches $7.6M over three years.

$2.08M/yr status-quo run cost • **$7.6M** 3-yr cost of inaction

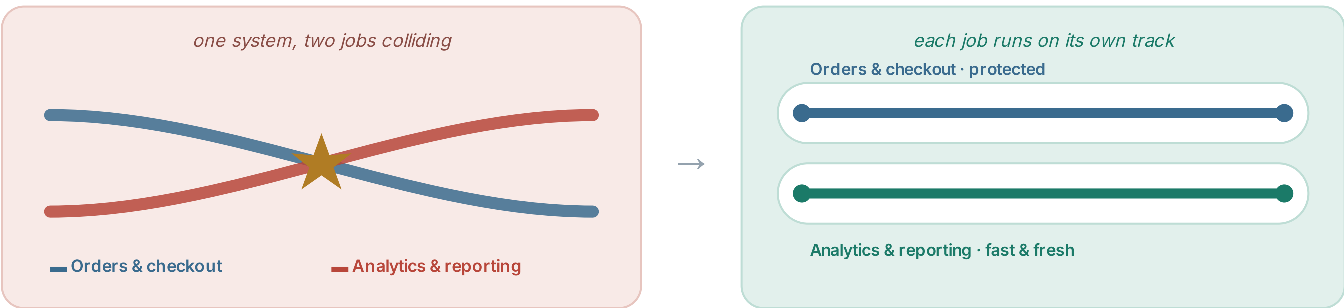
03 Where we are vs. where we need to be

outcome, not design

This describes a **property we want** — the two jobs no longer competing — not how to build it. The "after" picture is an outcome to verify, not an architecture to adopt.

TODAY — SHARED LANE

TARGET — SEPARATE LANES



Two jobs compete for the same resources. The store suffers whenever analytics runs hot.

Neither job can slow the other. The store is never degraded by a report; answers arrive in seconds.

FIG. 2 — THE SHIFT IS CONCEPTUAL: FROM CONTENTION TO ISOLATION. THE "HOW" IS INTENTIONALLY LEFT OPEN FOR PASS 2.

04 What this problem is — and isn't

scope boundary

Scope here means **which problems we commit to solve**, not which tools we pick. Several capabilities from the original proposal are real, but they are *separate* problems that ride on this foundation; folding them in now is what makes the work risky and the timeline slip.

✓ In scope — the problem we solve now
Stop analytical and transactional work from competing for the same resources
Protect the storefront so analytics can never degrade checkout at peak
Bring analytical query latency and data freshness into a range that supports timely decisions
Cut the operational toil and incident load this coupling creates
Make intraday pricing decisions possible — the highest-confidence value lever

× Out of scope — separate problems
Natural-language / self-service query interface a consumption-experience problem — evaluate after the foundation is proven
Autonomous / agentic incident remediation an ops-automation problem with its own risk profile
Real-time personalization engine depends on this foundation; its own initiative (later phase)
Advanced ML & predictive analytics; historical data beyond 12 months
Data governance & compliance program
Any specific technology choice — that is Pass 2

Why the cuts: the original $140K / 4-month / 3-engineer plan is credible for separation alone. The natural-language and agentic layers carry most of the execution risk and most of the soft, low-confidence revenue — they deserve their own justification, not a free ride on this one.

05 What ”done” must prove

verifiable acceptance

Each criterion is a pass/fail test tied to one of the program's own KPIs. "Done" is not "it feels faster" — it is these measurements, taken these ways.

ID	ACCEPTANCE CRITERION (VERIFIABLE)	HOW WE VERIFY IT	KPI IT PROVES
AC-1	**Isolation under peak.** With analytics running at full load during a simulated 75,000-orders/day peak, transactional performance shows no measurable degradation vs. the analytics-off baseline, and zero lock-waits are attributable to analytics.	Load test at peak volume; compare transactional commit latency analytics-on vs. analytics-off; lock-wait audit.	System availability 97.8% → 99.5%
AC-2	**Interactive latency.** The standard analytical query suite returns at p95 ≤ 5 s and p99 ≤ 15 s over a representative week.	Query-monitoring logs, measured across a full business week.	Query latency p95 12 min → <5 sec
AC-3	**Decision-grade freshness.** Median end-to-end lag from a change in the source to its availability for analysis is ≤ 5 minutes, sampled continuously.	Continuous freshness probe comparing source commit time to availability time.	Data freshness 24 hr → <5 min
AC-4	**Intraday pricing cadence.** Pricing inputs refresh and are queryable on a ≤ 15-minute cadence through a full peak window, demonstrated end to end.	Run the pricing job at 15-min cadence across a peak window; confirm inputs current each cycle.	Pricing frequency 1×/day → 15-min
AC-5	**Incident load falls.** Over a 90-day window after cutover, analytics-attributed incidents run at ≤ 0.5 per month.	Incident-tracking system, 90-day trailing count, analytics-tagged only.	Incidents / month 2.8 → <0.5

ID	ACCEPTANCE CRITERION (VERIFIABLE)	HOW WE VERIFY IT	KPI IT PROVES
AC-6	**Capacity reclaimed.** Data-team time on analytics maintenance falls to ≤ 30% of capacity, sustained across a quarter.	Time-tracking records, quarter-over-quarter, analytics-maintenance category.	Maintenance burden 60–70% → 20–30%

AC-1 is the gate that matters most and is easiest to skip: latency targets can be met while the storefront is still exposed at peak. Put AC-1 in the early go/no-go, not the final pilot.

06 Success metrics — current → target

the scoreboard

The full set of measures we will hold ourselves to, with the numbers as they stand today. Targets are the program's own 12-month figures.

MEASURE	TODAY		TARGET (12 MO)
Query latency · p95	12 min	→	<5 sec
Query latency · p99	28 min	→	<15 sec
Data freshness	24 hr	→	<5 min
System availability	97.8%	→	99.5%
Incidents per month	2.8	→	<0.5
Incident recovery (MTTR)	5.2 hr	→	30 min
Maintenance burden	60–70%	→	20–30%
Pricing optimization cadence	1×/day	→	15-min
New capabilities shipped	0–1/qtr	→	2–3/qtr
Business users querying data	5–10	→	30–40

Measurement sources are the existing query-monitoring, uptime, incident-tracking, and time-tracking systems — the baselines above already come from them, so the scoreboard is live from day one.

07 Open assumptions to close

before we commit

These are the beliefs the problem definition rests on. Each is a thing we should confirm — most before the build, a couple during validation — because if one is wrong, the shape of the work changes.

ID	ASSUMPTION	WHY IT MATTERS / IMPACT IF WRONG	HOW WE CLOSE IT
OA-1	The toil and incidents are mostly **caused by the workload coupling**, so separation removes them.	The whole cost-reduction case rides on this. If toil comes from elsewhere, separation won't move maintenance to 20–30%.	Root-cause the last 90 days of incidents; tag each as coupling-related or not.

ID	ASSUMPTION	WHY IT MATTERS / IMPACT IF WRONG	HOW WE CLOSE IT
OA-2	The **peak conversion loss is actually analytics-induced**, not another bottleneck.	If checkout slows for other reasons, isolation won't recover those sales and AC-1's value drops.	Correlate past peak slowdowns with analytics activity in the same windows.
OA-3	**75,000 orders/day is the right design point** for peak.	Sets the bar AC-1 must clear. Too low and we pass a test that doesn't reflect reality.	Confirm against actual peak-day logs from the last 12 months, plus growth headroom.
OA-4	Analytical data can be captured **without adding load to the store's critical path**.	If capture itself touches the live store, we reintroduce the very contention we're removing.	Prototype capture early against production-like volumes; measure source impact.
OA-5	**Pricing is the highest-confidence value lever** and a sound first proof point.	If pricing uplift is weak, the in-scope revenue case leans harder on softer items.	Validate the pricing logic against historical data; size the realistic uplift.
OA-6	The **60–70% maintenance figure is analytics-attributable**, not general team duties.	Defines the size of the prize. If much of it is unrelated work, the savings shrink.	Break down the existing time-tracking by activity before setting the target.

08 What we're asking the team to agree on

decision

Sign off on four things, and Pass 2 can start.

1. **The root cause** is workload contention — one database doing two competing jobs — not a tuning or hardware problem.
2. **The scope boundary** in §04: we solve separation, isolation, freshness, toil, and intraday pricing now; the natural-language and agentic layers are separate, later problems.
3. **"Done" is the six acceptance criteria** in §05, with AC-1 (peak isolation) as an early go/no-go gate rather than a final-pilot check.
4. **We will close the open assumptions** in §07 before committing the build — starting with incident root-causing and the peak conversion-loss correlation.

Owners for the calls above this team's pay grade: scope boundary → CTO (delivery & risk); KPI targets and the maintenance-burden number → VP Data; the peak-isolation acceptance definition → Data Engineering Lead to draft, VP Data to ratify. Pass 2 proposes the solution and technology against exactly this problem.